1. An image contains high-frequency noise. Explain and justify the appropriate frequency domain filter, including its working and impact on image quality.

Answer:

- **Filter Used:** Low-Pass Filter (LPF), preferably Gaussian Low-Pass Filter.
- **Working:** It allows low-frequency components to pass and blocks high-frequency components where noise is present.
- **Impact on Image Quality:**
  - Reduces high-frequency noise.
  - Produces a smoother image.
  - Slightly blurs edges and fine details.
- **Justification:** Since noise mainly exists in high frequencies, LPF effectively removes it while preserving the main image content.

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2. You need to preserve low-frequency components while removing noise. Describe and apply a suitable filtering technique.

Answer:

- **Technique:** Gaussian Low-Pass Filter.
- **Working:**
  1. Convert the image to the frequency domain using FFT.
  2. Apply the Gaussian LPF.
  3. Perform the inverse FFT to obtain the filtered image.
- **Result:** Low-frequency information is preserved while high-frequency noise is removed, producing a smooth and clear image.

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3. An image requires edge enhancement in the frequency domain. Explain and justify the filtering approach used.

Answer:

- **Filter Used:** High-Pass Filter (HPF).
- **Working:** Blocks low-frequency components and passes high-frequency components containing edges and fine details.
- **Effect:**

- Sharpens edges.
- Enhances image details.
- May also increase noise.

- **Justification:** Edges are represented by high-frequency components, so HPF enhances them effectively.

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4. A smooth version of an image is required. Describe and justify the frequency domain technique used for smoothing.

Answer:

- **Technique:** Ideal, Butterworth, or Gaussian Low-Pass Filter.
- **Working:** Removes high-frequency components while retaining low-frequency information.
- **Result:**
    - Smooth image.
    - Reduced noise.
    - Slight edge blurring.
- **Justification:** LPF is the standard technique for image smoothing.

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5. An object is clearly brighter than the background. Explain and justify the segmentation technique used.

Answer:

- **Technique:** Threshold Segmentation.
- **Working:** Select a threshold value (T). Pixels greater than T become the object, while the rest become the background.
- **Effectiveness:**
    - Simple.
    - Fast.
    - Highly accurate when object and background intensities differ significantly.

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6. You need to separate foreground and background based on intensity. Describe and apply an appropriate segmentation method.

Answer:

- **Method:** Thresholding (Global or Otsu's Thresholding).
- **Process:**
    1. Compute the threshold.
    2. Compare each pixel intensity with the threshold.
    3. Classify pixels into foreground and background.
- **Result:** Produces a binary image separating the object from the background.

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7. An image requires identification of object edges. Explain and justify a suitable edge detection method.

Answer:

- **Method:** Canny Edge Detection.
- **Working:**
    1. Smooth the image using a Gaussian filter.
    2. Compute gradients.
    3. Apply Non-Maximum Suppression.
    4. Perform Double Thresholding.
    5. Track edges using Hysteresis.
- **Effectiveness:**
    - Detects accurate edges.
    - Removes noise.
    - Produces continuous edges.

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8. Adjacent pixels have similar intensity values. Describe and apply a segmentation technique to group such pixels.

Answer:

- **Technique:** Region Growing.
- **Working:**
    1. Select seed pixels.
    2. Compare neighboring pixel intensities.

3. Add similar pixels to the region.
4. Continue until no similar neighbors remain.

- **Reasoning:** Groups connected pixels with similar intensity into meaningful regions.

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9. Detected edges in an image appear broken. Explain and justify a method to improve edge continuity.

Answer:

- **Method:** Morphological Closing (Dilation followed by Erosion).
- **Working:**
  - Dilation connects nearby edge segments.
  - Erosion restores the original edge thickness.
- **Role in Post-Processing:**
  - Connects broken edges.
  - Improves edge continuity.
  - Produces smoother object boundaries.

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10. You need to extract object outlines from an image. Describe and apply an appropriate technique.

Answer:

- **Technique:** Boundary Extraction using Morphological Operations.
- **Working:**
  1. Erode the binary image.
  2. Subtract the eroded image from the original image.
- **Formula:**
  Boundary = Original Image − Eroded Image
- **Result:** Extracts only the object's outer boundary accurately.